TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – Computer Organization & Architecture

Lecture No.- 10

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Topics to be Covered

1:- Storage Device – Primary Memory
2:- Cache Memory
3:- Software and its type
4:- System Software
5:- Application Software
6:- Difference between system and Application Software

Advantages of ROM

- **Non-Volatile** – Retains data without power.
- **Security** – Prevents unauthorized changes.
- **Reliable** – Data remains intact over time.
- **Cost-Effective** – Cheap for large-scale production.
- **Fast Access** – Quick retrieval of stored data.

प्राप्त कर सकते है।

[Decoder + OR] → ROM

Disadvantages of ROM

- **Limited Modifiability** – Data can't be easily changed.
- **Low Storage Capacity** – Not suitable for large data storage.
- **Slow Write Speeds** – Writing data is slow.
- **Physical Wear** – Can wear out after many write cycles (EEPROM/EPROM).
- **High Initial Cost** – Expensive to manufacture in small quantities (Mask ROM).

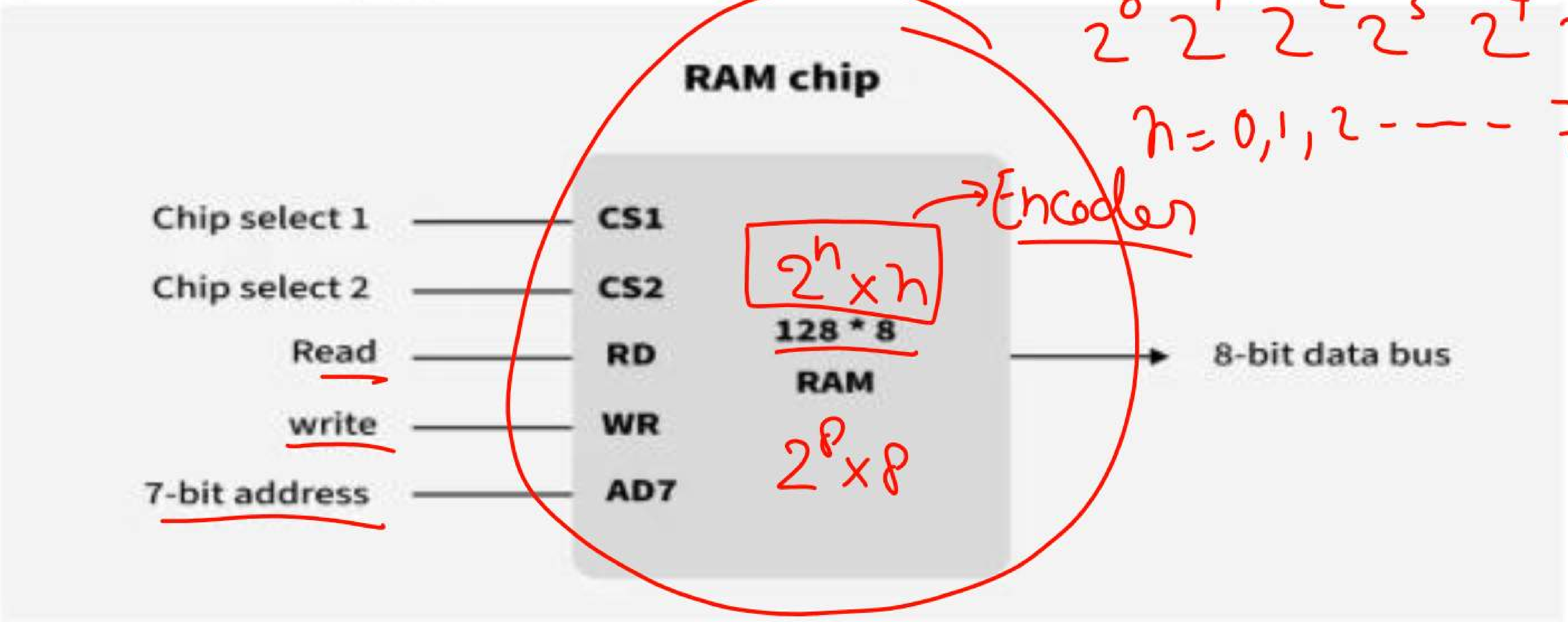
Topic: Primary Memory- RAM

What is RAM?

Random Access Memory, is a type of computer memory that allows data to be read and written randomly, meaning that the computer can access any location in the memory directly rather than having to read the data in a specific order. This makes RAM an essential component of a computer system, as it enables the CPU to access data quickly and efficiently.

RAM is volatile in nature, which means if the power goes off, the stored information is lost. RAM is used to store the data that is currently processed by the CPU. Most of the programs and data that are modifiable are stored in RAM.

The block diagram of the RAM chip is given below:



$2^0\ 2^1\ 2^2\ 2^3\ 2^4\ 2^5\ 2^6\ 2^7$

$n = 0, 1, 2 - - - 7$

Encoder

$\boxed{2^n \times n}$

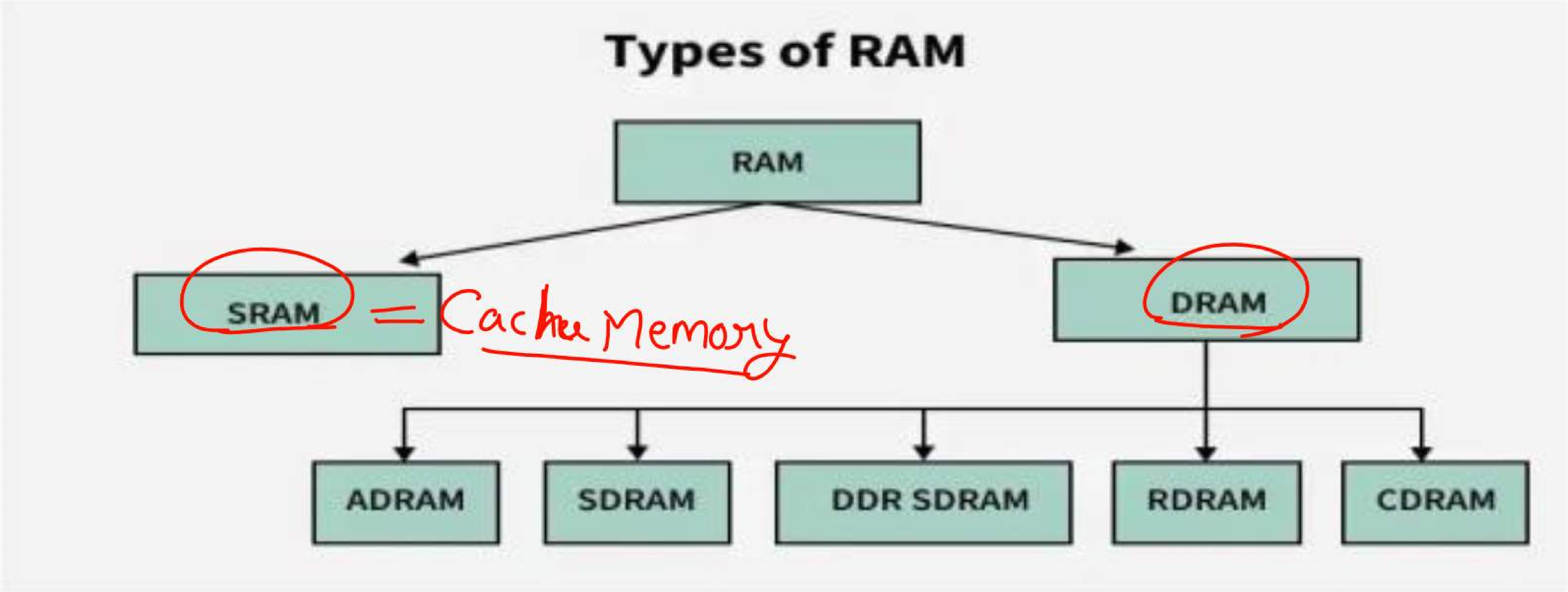
$2^8 \times 8$

Topic: Primary Memory- RAM

Types of RAM?

Mainly RAM have 2types

- SRAM (Static RAM)
- DRAM (Dynamic RAM)



Bus

Data Bus | Address Bus | Control Bus

Clock signal / Control signal

Control Bus → 8 bit

Data Bus → Only Data

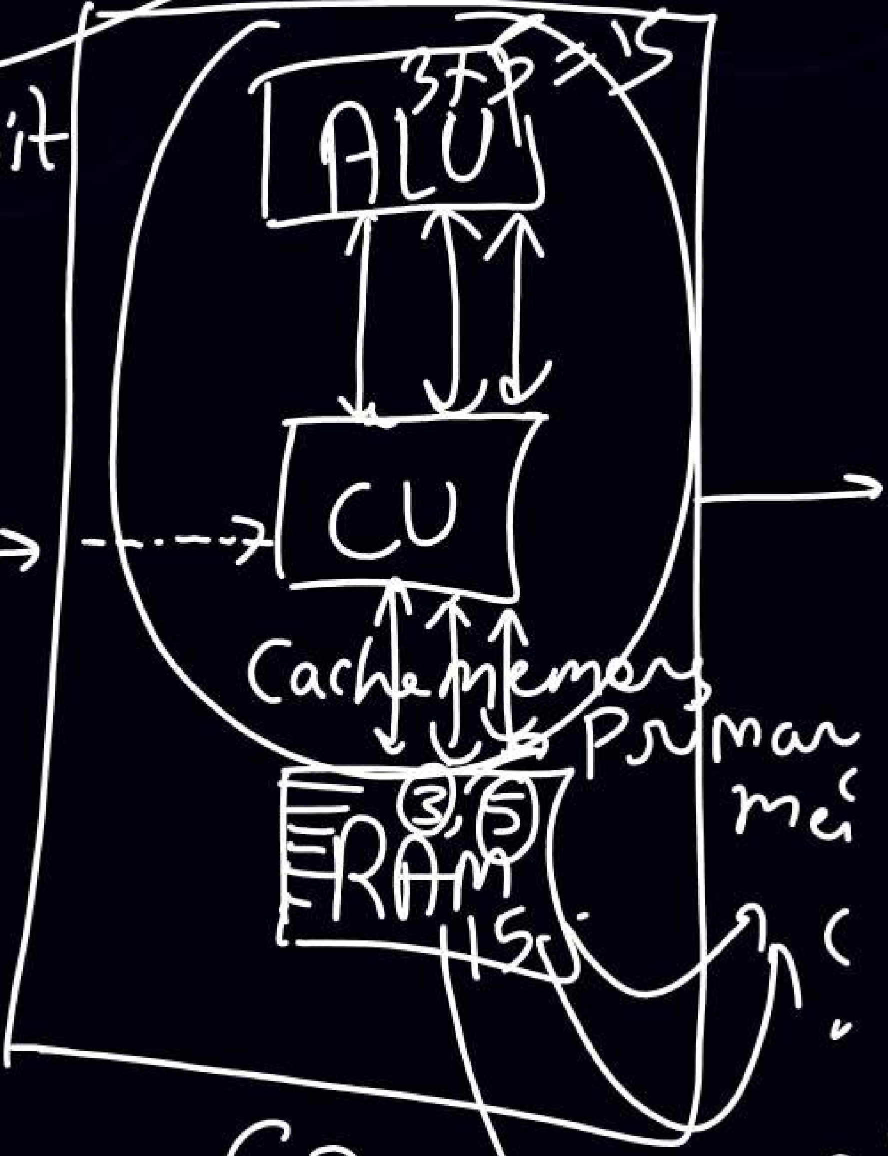
$\boxed{8\text{-bit}}$

Address Bus → It hold Address of Input, Output, Intermediate Data Address

(8 bit / 16 bit)

Max Place

Input 3x5 →



CPU

DRAM (Capacitor)

Cache Memory (SRAM)

Types of RAM

RAM is further divided into two types, SRAM - Static Random Access Memory and DRAM- Dynamic Random Access Memory. Let's learn about both of these types in more detail.

1. SRAM (Static Random Access memory)

SRAM is used for Cache memory, it can hold the data as long as the power availability is there. It is refreshed simultaneously to store the present information. It is made with CMOS technology. It contains 4 to 6 transistors and it also uses clocks. It does not require a periodic refresh cycle due to the presence of transistors. Although SRAM is faster, it requires more power and is more expensive. Since SRAM requires more power, more heat is lost here as well, another drawback of SRAM is that it can not store more bits per chip, for instance, for the same amount of memory stored in DRAM, SRAM would require one more chip.

Function of SRAM

The function of SRAM is that it provides a direct interface with the Central Processing Unit at higher speeds.

Characteristics of SRAM

- SRAM is used as the Cache memory inside the computer.
- SRAM is known to be the fastest among all memories.
- SRAM is costlier.
- SRAM has a lower density (number of memory cells per unit area).
- The power consumption of SRAM is less but when it is operated at higher frequencies, the power consumption of SRAM is compatible with DRAM.

2. DRAM (Dynamic Random Access memory)

DRAM is used for the Main memory, it has a different construction than SRAM, it uses one transistor and one capacitor (also known as a conductor), which is needed to get recharged in milliseconds due to the presence of the capacitor. Dynamic RAM was the first sold memory integrated circuit. DRAM is the second most compact technology in production (the First is Flash Memory). DRAM has one transistor and one capacitor in 1 memory bit. Although DRAM is slower, it can store more bits per chip, for instance, for the same amount of memory stored in SRAM, DRAM requires one less chip. DRAM requires less power and hence, less heat is produced.

click → charge

Capacitor electronic component

Milli second

discharge

Storage of charge

10^{-3} Second

Function of DRAM

The function of DRAM is that it is used for programming code by a computer processor to function. It is used in our PCs (Personal Computers).

Characteristics of DRAM

- DRAM is used as the Main Memory inside the computer.
- DRAM is known to be a fast memory but not as fast as SRAM.
- DRAM is cheaper as compared to SRAM.
- DRAM has a higher density (number of memory cells per unit area)
- The power consumption by DRAM is more

Difference Between SRAM and DRAM

Feature	SRAM (Static RAM)	DRAM (Dynamic RAM)
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Power Consumption	Requires **more power**	Requires **less power**
Cost	**More expensive**	**Less expensive**
Speed	**Faster** due to no need for refreshing	**Slower** because it needs to be refreshed
Usage	Used in **cache memory** for quick access	Used in **main memory** for large data storage

CMOS

ROM vs RAM

BIOS
firmware

ROM	RAM
Non-volatile memory used for permanent storage	Volatile memory used for temporary storage
Generally slower than RAM	High-speed access
Primarily read-only	Read and write operations
ROM can hold more than just megabytes	RAM can be stored in gigabytes
Data accessible is less difficult but more restrictive	Data accessibility is easy
Cheaper than RAM	High cost as compared to ROM
Used for the permanent storage of data	Used for temporary storage of data

Advantages of RAM

- **Speed**: RAM is faster than other types of storage like ROM, hard drives or SSDs, allowing for quick access to data and smooth performance of applications.
- **Multitasking**: More RAM allows a computer to handle multiple applications simultaneously without slowing down.
- **Flexibility**: RAM can be easily upgraded, enhancing a computer's performance and extending its usability.
- **Volatile Storage**: RAM automatically clears its data when the computer is turned off, reducing the risk of unwanted data accumulation.

Disadvantages of RAM

- **Volatility**: Data stored in RAM is lost when the computer is turned off, which means important data must be saved to permanent storage.
- **Cost**: RAM can be more expensive per gigabyte compared to other storage options like hard drives or SSDs.
- **Limited Storage**: RAM has a limited capacity, so it cannot store large amounts of data permanently.
- **Power Consumption**: RAM requires continuous power to retain data, contributing to the overall power consumption of the device.
- **Physical Space**: Increasing RAM requires physical space in the computer, which might be limited to smaller devices like laptops and tablets.

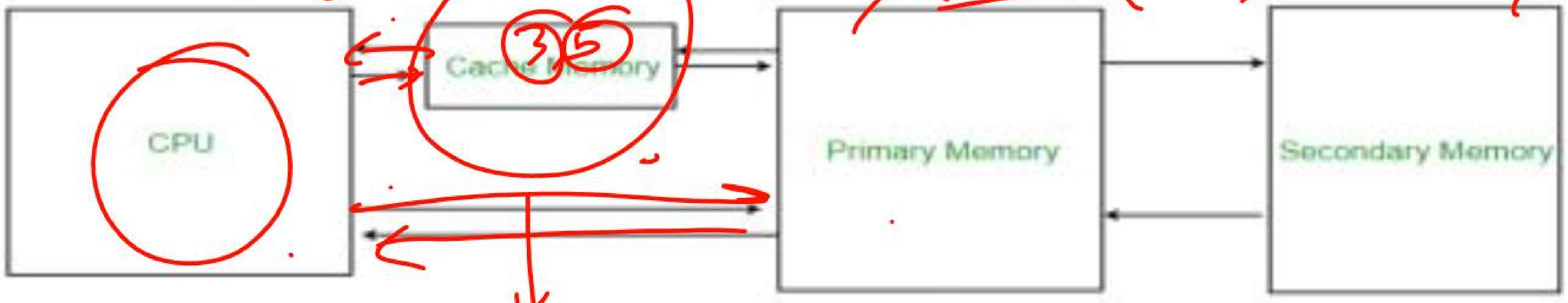
Characteristics of Cache Memory

1-bit data

Register > Cache Memory > RAM

- Extremely fast memory type that acts as a buffer between RAM and the CPU.
- Holds frequently requested data and instructions, ensuring that they are immediately available to the CPU when needed.
- Costlier than main memory or disk memory but more economical than CPU registers.
- Used to speed up processing and synchronize with the high-speed CPU.

RAM (GB)



Cache Memory

Levels of Memory

Register

Program Counter

→ next Address hold

- **Level 1 or Register:** It is a type of memory in which data is stored and accepted that are immediately stored in the CPU. The most commonly used register is Accumulator, Program counter, Address Register, etc.
- **Level 2 or Cache memory:** It is the fastest memory that has faster access time where data is temporarily stored for faster access.
- **Level 3 or Main Memory:** It is the memory on which the computer works currently. It is small in size and once power is off data no longer stays in this memory.
- **Level 4 or Secondary Memory:** It is external memory that is not as fast as the main memory but data stays permanently in this memory.

Advantages

- Cache Memory is faster in comparison to main memory and secondary memory.
- Programs stored by Cache Memory can be executed in less time.
- The data access time of Cache Memory is less than that of the main memory.
- Cache Memory stored data and instructions that are regularly used by the CPU, therefore it increases the performance of the CPU.

Disadvantages

- Cache Memory is costlier than primary memory and secondary memory .
- Data is stored on a temporary basis in Cache Memory. voltile memory
- Whenever the system is turned off, data and instructions stored in cache memory get destroyed.
- The high cost of cache memory increases the price of the Computer System.

Topic: Memory Units

$2^0\ 2^1\ 2^2\ 2^3\ 2^4$ ....

$10^3 = 1000 \simeq 1024$

1	**Bit (Binary Digit)** A binary digit is logical 0 and 1 representing a passive or an active state of a component in an electric circuit.
2	**Nibble** A group of 4 bits is called nibble.
3	**Byte** A group of 8 bits is called byte. A byte is the smallest unit which can represent a data item or a character.
4	**Word** A computer word, like a byte, is a group of fixed number of bits processed as a unit, which varies from computer to computer but is fixed for each computer. The length of a computer word is called word-size or word length. It may be as small as 8 bits or may be as long as 96 bits. A computer stores the information in the form of computer words.

2^{10}

$2^{10} = 1024$

$2^9 = 512$

4bits → 1 Nibble

8bits → 1 Byte

0

1

word → Byte

S = "RAHUL" → 1 byte

Print(Size(S))

5 byte

1KB = 1024 Byte

$= 10^3$ Bytes

1MB = 10^6 Byte

$= 1024 \times 1024$ Byte

1	**Kilobyte (KB)** 1 KB = 1024 Bytes
2	**Megabyte (MB)** 1 MB = 1024 KB
3	**GigaByte (GB)** 1 GB = 1024 MB
4	**TeraByte (TB)** 1 TB = 1024 GB
5	**PetaByte (PB)** 1 PB = 1024 TB

1KB = 1024 Bytes $= 10^3$

1MB = 1024KB

1MB = 10^6 Byte

1GB = 10^9 Bytes

1TB = 10^{12} Bytes

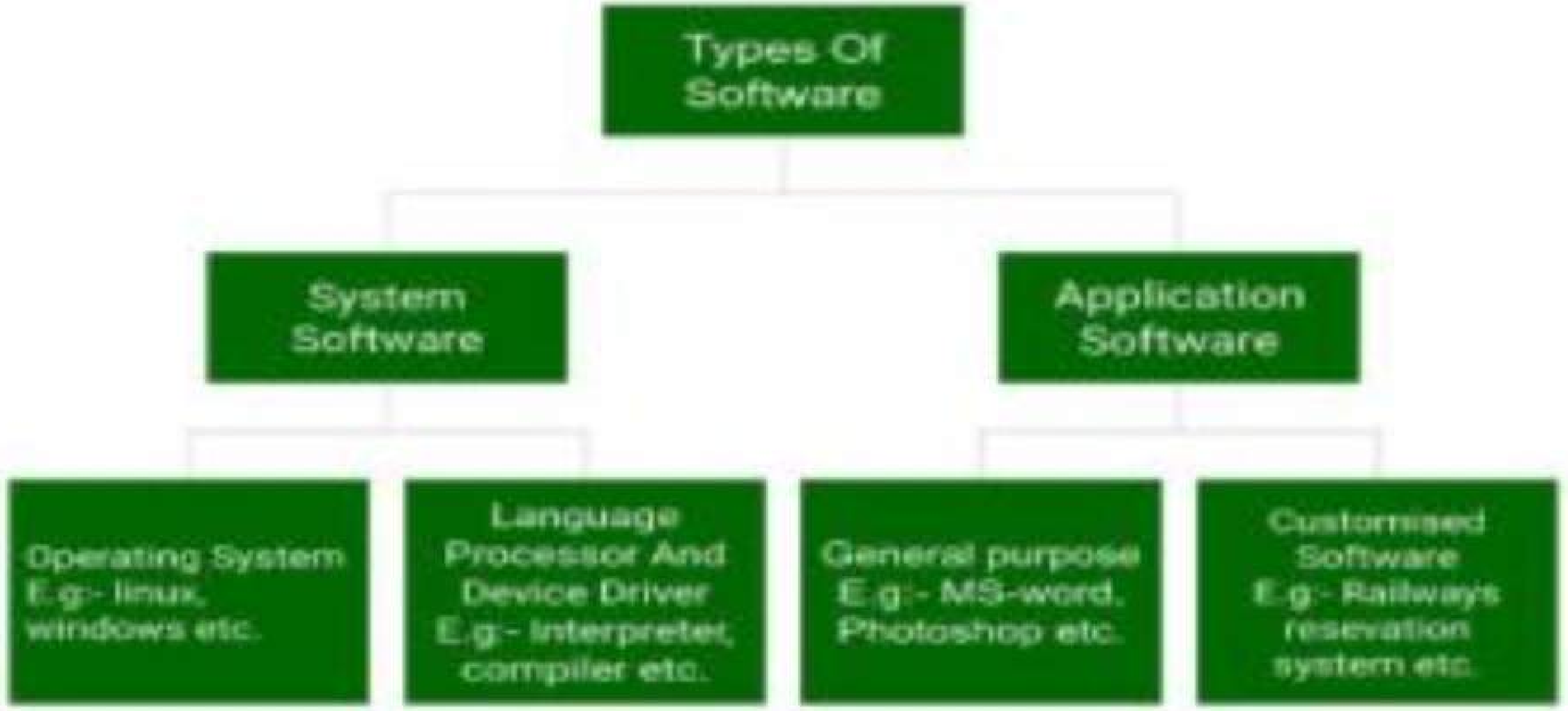
1PB = 10^3 TB

In a computer system, the software is basically a set of instructions or commands that tell a computer to do a specific task that serves its users.

- A software is not a physical thing like hardware, it rather makes the hardware work as per user requirements by giving instructions.
- Examples of software are MS-Word, MS-Excel, PowerPoint and Web Browser

Types of Software

The chart below describes the types of software:



1. **System Software**
   - Operating System (like Windows and Linux)
   - Language Processor
   - Device Driver

2. **Application Software**
   - General Purpose Software
   - Customize Software
   - Utility Software

System Software

System software is software that directly operates the computer hardware and provides the basic functionality to the users as well as to the other software to operate smoothly.

- System software basically controls a computer's internal functioning and also controls hardware devices such as monitors, printers, and storage devices, etc.
- It is like an interface between hardware and user applications, it helps them to communicate with each other because hardware understands machine language(i.e. 1 or 0) whereas user applications are work in human-readable languages like English, Hindi, German, etc.

Types of System Software

It has two subtypes which are:

1. **Operating System:** It is the main program of a computer system. When the computer system ON it is the first software that loads into the computer's memory. Basically, it manages all the resources such as computer memory, CPU, printer, hard disk, etc., and provides an interface to the user, which helps the user to interact with the computer system.
2. **Language Processor:** As we know that system software converts the human-readable language into a machine language and vice versa. So, the conversion is done by the language processor. It converts programs written in high-level programming languages like Java, C, C++, Python, etc(known as source code), into sets of instructions that are easily readable by machines(known as object code or machine code).
3. **Device Driver:** A device driver is a program or software that controls a device and helps that device to perform its functions. Every device like a printer, mouse, modem, etc. needs a driver to connect with the computer system eternally. So, when you connect a new device with your computer system, first you need to install the driver of that device so that your operating system knows how to control or manage that device.

Application Software

Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as application software.

- Application software is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements.
- Examples include word processors, spreadsheets, database management, inventory, payroll programs, etc.

Types of Application Software

There are different types of application software and those are:

1. **General Purpose Software:** This type of application software is used for a variety of tasks and it is not limited to performing a specific task only. For example, MS-Word, MS-Excel, PowerPoint, etc.
2. **Customized Software:** This type of application software is used or designed to perform specific tasks or functions or designed for specific organizations. For example, railway reservation system, airline reservation system, invoice management system, etc.
3. **Utility Software:** This type of application software is used to support the computer infrastructure. It is designed to analyze, configure, optimize and maintains the system, and take care of its requirements as well. For example, antivirus, disk fragmenter, memory tester, disk repair, disk cleaners, registry cleaners, disk space analyzer, etc.

System Software	Application Software
It is designed to manage the resources of the computer system, like memory and process management, etc.	It is designed to fulfill the requirements of the user for performing specific tasks.
Written in a low-level language.	Written in a high-level language.
Less interactive for the users.	More interactive for the users.
System software plays vital role for the effective functioning of a system.	Application software is not so important for the functioning of the system, as it is task specific.
It is independent of the application software to run.	It needs system software to run.

THANK YOU